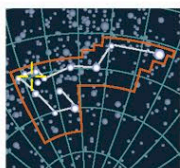


EMISSION NEBULA

Carina Nebula



CATALOG NUMBER
NGC 3372

DISTANCE FROM SUN
8,000 light-years

MAGNITUDE 1

CARINA

Also known as the Eta (η) Carinae Nebula, this is one of the largest and brightest nebulae to be discovered. It has a diameter of more than 200 light-years, stretching up to 300 light-years if its fainter outer filaments are included. Within its heart, and heating up its dust and gas, is an interesting zoo of young stars. These include examples of the most massive stars known, with a spectral type of O3 (see pp.232–233). This type of star was first discovered in the Carina Nebula, and the nebula remains the closest location of O3 stars to Earth. Also within the Carina Nebula are three Wolf-Rayet stars with spectral type WN (see pp.254–255). These stars are believed to be evolved O3 stars with very large rates of mass ejection. One of the best-known features within the Carina Nebula is the blue supergiant star Eta (η) Carinae (see p.262), embedded within part of the nebula known as the Keyhole Nebula. Recent observations made with infrared



PROBING THE NEBULA

An infrared image reveals the stars lying within the nebula's dense dust and gas. The open clusters Trumpler 14 and Trumpler 16 are visible to the left and top of the image.

telescopes reveal that portions of the Carina Nebula are moving at very high speeds—up to 522,000 mph (828,000 kph)—in varying directions. Collisions of interstellar clouds at these speeds heat material to such high temperatures that it emits high-energy X-rays, and the entire Carina Nebula is a source of extended X-ray emission. The movement of these clouds of material is thought to be due to the strong stellar winds emitted by the massive stars within, bombarding the surrounding material and accelerating it to its high velocities.

ERODING TOWER

A tower of cool hydrogen gas and dust 3 light-years long extends from the Carina Nebula in this false-color Hubble image. The tower is being eroded by the energy from hot, young stars nearby.



EMISSION NEBULA

RCW 49



CATALOG NUMBERS
RCW 49, GUM 29

DISTANCE FROM SUN
14,000 light-years

CARINA

One of the most productive regions of star formation to have been found in the Milky Way, RCW 49 spans a distance of about 350 light-years. It is thought that over 2,200 stars reside within RCW 49, but because of the nebula's dense areas of dust and gas, the stars are hidden from view at optical wavelengths of light. However, the infrared telescope onboard the Spitzer spacecraft (see panel, right) has recently revealed the presence of up to 300 newly formed stars. Stars have been observed at every stage of their early evolution in this area, making it a remarkable source of data for studying star formation and development. One surprising preliminary observation suggests that most of the stars have accretion disks around them. This is a far higher ratio than would usually be expected. Detailed observations of two of the disks reveal that they are composed of exactly what is required in a planet-forming system. These are the farthest and faintest potential planet-forming disks ever observed. This discovery supports the theory that planet-forming disks are a natural part of a star's evolution. It also suggests that solar systems like our own are probably not rare in the Milky Way (see pp.296–299).

COSMIC CONSTRUCTION

This false-color image, composed of four separate images taken in different infrared wavelengths, reveals more than 300 newborn stars scattered throughout the RCW 49 nebula. The oldest stars of the nebula appear in the center in blue, gas filaments appear in green, and dusty tendrils are shown in pink.

EXPLORING SPACE

SPITZER TELESCOPE

Launched in August 2003, the Spitzer Telescope is one of the largest infrared telescopes put into orbit. It has been very successful in probing the dense dust and gas that lies in the interstellar medium and has revealed features and details within star-forming clouds that have never been seen before. As Spitzer observes in infrared, its instruments are cooled almost to absolute zero to ensure that their own heat does not interfere with the observations. A solar shield protects the telescope from the Sun.



INSIDE SPITZER
The Spitzer craft has a 34-in (85-cm) telescope and three super-cooled processing instruments.